

Near-Optimal Lower Bounds for Spectrum Estimation

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Abstract

This PDF contains an LLM-generated proof of an $\Omega\left(\frac{d^2}{\epsilon^2 \log^2 d}\right)$ copy complexity lower bound for estimating the spectrum of an unknown d -dimensional quantum state to sufficiently small sorted TV-distance ϵ . This result was obtained on the 26th of July. The proof ideas and all contents of the present PDF (except this abstract) were generated entirely by ChatGPT 5.6 given access to an early draft of our work arXiv:2607.29680. The human authors are in the process of writing up a more digestible version of this proof, which will be uploaded to arXiv soon.

1 Introduction

For a matrix $X \in \mathbb{C}^{d \times d}$, write

$$\mathrm{tr} X := \frac{1}{d} \mathrm{Tr} X$$

for normalized trace. If $p, q \in \mathbb{R}^d$ are probability vectors, with nonincreasing rearrangements $p^\downarrow, q^\downarrow$, define

$$d_{\mathrm{TV}}^\downarrow(p, q) := \frac{1}{2} \sum_{i=1}^d |p_i^\downarrow - q_i^\downarrow|.$$

For density matrices ρ, σ , let

$$D_{\mathrm{tr}}(\rho, \sigma) := \frac{1}{2} \|\rho - \sigma\|_1.$$

Let $N_{\mathrm{spec}}(d, \varepsilon)$ be the least number of copies that suffice to estimate the spectrum of every d -dimensional state to sorted-TV error at most ε , with success probability at least $2/3$.

Theorem 1.1 (Main theorem). *There are absolute constants $c, C > 0$ such that, for every sufficiently large d and every*

$$0 < \varepsilon \leq \frac{c}{\log d},$$

one has

$$N_{\mathrm{spec}}(d, \varepsilon) \geq C \frac{d^2}{\varepsilon^2 \log^2 d}.$$

The conclusion remains valid when the right-hand side exceeds d^2 .

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1.1 The proof in three equations

The quantitative conclusion is forced by three scales:

$$\text{spectral separation} \asymp \frac{\eta}{k}, \quad \text{nonconstant log-likelihood} \lesssim d^2 \theta^k, \quad \frac{\eta \sqrt{N}}{d} \asymp 1. \quad (1)$$

The first relation comes from a degree- k Chebyshev alternation pair and a Gibbs perturbation of size η . The second comes from an order- k zero of the HCIZ free-energy difference. The third keeps the HCIZ coupling inside a fixed zero-free disc. Thus

$$k \asymp \log d, \quad \eta \asymp k\varepsilon, \quad N \asymp \frac{d^2}{\eta^2} \asymp \frac{d^2}{\varepsilon^2 \log^2 d}.$$

The proof below is devoted to making these three relations uniform over the random Young diagram observed by the learner.

1.2 The underlying mechanism

Haar randomization removes the labels of the eigenvalues. After averaging over the eigenbasis, the experiment is described by a Young diagram λ , and its likelihood is a unitarily invariant function of the spectrum. For fixed B , the order- m coefficient of the HCIZ free energy is a conjugation-invariant homogeneous polynomial of degree m in the first external field. Restricted to its eigenvalues, it is a symmetric polynomial of degree m , and hence depends only on the first m power traces. Therefore

$$\text{tr}(D_a^j) = \text{tr}(D_b^j), \quad j < k,$$

directly forces equality of the first $k - 1$ free-energy coefficients.

The logarithm is the right object because it is the additive log-likelihood and because stable nonvanishing permits a dimension-uniform analytic bound on it. Novak proves that the HCIZ integral has no zeros on a dimension-independent small disc. This gives a single analytic branch of the free energy

$$\Phi_d(A, B; z) := d^{-2} \log I_d(A, B; z).$$

The elementary estimate $|I_d(A, B; z)| \leq e^{|z|d^2}$ bounds the real part of this logarithm; a half-disc lemma then bounds its full modulus. Finally, Schwarz's lemma turns the order- k zero into an $O(d^2 \theta^k)$ pointwise estimate.

One may still interpret the Taylor coefficients of $\log I_d$ as connected invariants or cumulants. That interpretation explains the free-energy language, but it is not needed in the proof.

1.3 Classical perspective

For the classical problem of learning a distribution up to permutation under sorted ℓ_1 loss, Han, Jiao, and Weissman prove, for $n \gtrsim d/\log d$, the minimax risk

$$\sqrt{\frac{d}{n \log n}} + \tilde{\Theta} \left(\sqrt{\frac{d}{n}} \wedge n^{-1/3} \right). \quad (2)$$

The first term is a moment-matching or bias term; the second is a variance term with its own phase transition [1, Theorem 1]. The present theorem identifies a quantum analogue of the first mechanism. It should not be read as evidence that the full quantum minimax rate must be a single expression d^2/ε^2 .

1.4 Reading guide

The proof has one genuinely deep external input: Novak’s stable nonvanishing theorem for the HCIZ integral. Everything surrounding it is elementary or standard once the right representation is found.

Sections 2 and 3 contain the dimension reduction, hard-pair construction, and Schur–Weyl bookkeeping, and are kept compact. Section 4 derives an exact character–HCIZ *ratio*; its only delicate point is normalization. Section 5 is the analytic core, but the proof is now short: a direct invariant-polynomial lemma for the free-energy coefficients, Novak’s zero-free disc, an elementary half-disc bound, and Schwarz’s lemma. Section 6 proves the auxiliary Young-diagram window using only second moments. Section 7 normalizes the nearly constant likelihood and completes the testing reduction.

The two principal design choices remain endpoint-dependent Poissonization and the use of the HCIZ free energy. The first is an exact algebraic cancellation. The second identifies the analytic object whose low-order jet is controlled by moments and whose total size is only $O(d^2)$.

2 The hard spectra

We first remove an inessential divisibility nuisance.

Lemma 2.1 (Monotonicity in the dimension). *If $1 \leq d \leq D$, then*

$$N_{\text{spec}}(D, \varepsilon) \geq N_{\text{spec}}(d, \varepsilon).$$

Proof. Embed a d -dimensional state as $\rho \oplus 0_{D-d}$. Suppose a D -dimensional estimator outputs a probability vector q , which we may sort nonincreasingly. If p is the sorted spectrum of ρ , define the d -vector

$$\tilde{q} := \left(q_1, \dots, q_{d-1}, \sum_{i=d}^D q_i \right)$$

and then sort it. Before the final sorting,

$$\begin{aligned} \|\tilde{q} - p\|_1 &\leq \sum_{i=1}^{d-1} |q_i - p_i| + |q_d - p_d| + \sum_{i=d+1}^D q_i \\ &= \|q - (p, 0, \dots, 0)\|_1. \end{aligned}$$

Sorting can only improve the optimal coordinate matching with the sorted vector p . Thus any estimator in dimension D yields one with no larger sorted-TV error in dimension d . $\square$

All constructions below are carried out in a dimension d divisible by the even moment-matching degree k . In the final section, given an arbitrary ambient dimension D , we take

$$d = k \left\lfloor \frac{D}{k} \right\rfloor$$

and invoke Theorem 2.1. Since $k = O(\log D)$, this loses only an absolute constant in the final bound.

2.1 An extremal moment-matching pair

Fix an even integer $k \geq 2$. Let

$$x_j := \cos \frac{j\pi}{k}, \quad 0 \leq j \leq k,$$

and let $c_0 = c_k = 1/2$ and $c_j = 1$ for $1 \leq j \leq k-1$. Define

$$\mu_a := \frac{2}{k} \sum_{\substack{0 \leq j \leq k \\ j \text{ even}}} c_j \delta_{x_j}, \quad \mu_b := \frac{2}{k} \sum_{\substack{0 \leq j \leq k \\ j \text{ odd}}} c_j \delta_{x_j}. \quad (3)$$

Both are symmetric probability measures on $[-1, 1]$.

Proposition 2.2 (Chebyshev alternation pair). *The measures in (3) satisfy*

$$\int x^r d\mu_a(x) = \int x^r d\mu_b(x), \quad 0 \leq r < k,$$

and

$$W_1(\mu_a, \mu_b) = \frac{2}{k}.$$

More generally, for every increasing C^1 map $g : [-1, 1] \rightarrow \mathbb{R}$,

$$W_1(g_{\#}\mu_a, g_{\#}\mu_b) = \frac{g(1) - g(-1)}{k}. \quad (4)$$

Proof. For $1 \leq \ell \leq 2k-1$, pairing terms in the geometric sum of the $2k$ th roots of unity gives the Lobatto identity

$$\sum_{j=0}^k c_j \cos \frac{\ell j \pi}{k} = 0.$$

Every polynomial p of degree less than k has $p(\cos \theta)$ in the span of $1, \cos \theta, \dots, \cos((k-1)\theta)$. Since $(-1)^j = \cos(kj\pi/k)$, the product-to-sum identity and the preceding Lobatto identity yield

$$\sum_{j=0}^k c_j (-1)^j p(x_j) = 0.$$

Taking $p(x) = x^r$ proves moment matching.

For the Wasserstein statement, read the nodes from $-1 = x_k$ to $1 = x_0$. The signed masses of $\mu_a - \mu_b$ alternate as

$$\frac{1}{k}, -\frac{2}{k}, \frac{2}{k}, \dots, -\frac{2}{k}, \frac{1}{k}.$$

Thus the absolute cumulative signed mass is exactly $1/k$ between every two consecutive nodes. The one-dimensional identity

$$W_1(\alpha, \beta) = \int_{\mathbb{R}} |F_{\alpha} - F_{\beta}|$$

gives

$$W_1(\mu_a, \mu_b) = \frac{1}{k} \sum_{j=0}^{k-1} (x_j - x_{j+1}) = \frac{2}{k}.$$

After the increasing change of variables $y = g(x)$, the same calculation gives $k^{-1} \int_{-1}^1 g'(x) dx$, which is (4). $\square$

2.2 Gibbs transformation

Assume from now on that k divides d . Realize μ_a, μ_b exactly as the empirical eigenvalue measures of diagonal matrices $D_a, D_b \in \mathbb{C}^{d \times d}$. Their symmetry makes D_a, D_b traceless Hermitian contractions, and their normalized power traces agree below degree k .

Fix $\eta > 0$ and define

$$A_e := e^{\eta D_e}, \quad M_e := \text{tr } A_e = \int e^{\eta x} d\mu_e(x), \quad \sigma_e := \frac{A_e}{dM_e}. \quad (5)$$

The hard prior at endpoint e is a Haar-random conjugate of σ_e .

Proposition 2.3 (Gibbs separation). *There are absolute constants $c_0, \eta_{\text{sep}} > 0$ such that, whenever $0 < \eta \leq \eta_{\text{sep}}$,*

$$d_{\text{TV}}^\downarrow(\text{spec } \sigma_a, \text{spec } \sigma_b) \geq c_0 \frac{\eta}{k}. \quad (6)$$

Moreover,

$$e^{-\eta_{\text{sep}}} \leq M_e \leq e^{\eta_{\text{sep}}}. \quad (7)$$

Proof. For a probability vector p , let $\nu_p = d^{-1} \sum_i \delta_{dp_i}$. Monotone matching of equally weighted measures on the line gives

$$d_{\text{TV}}^\downarrow(p, q) = \frac{1}{2} W_1(\nu_p, \nu_q). \quad (8)$$

Define $g_e(x) := e^{\eta x}/M_e$. The empirical measures of $d \text{spec}(\sigma_e)$ are $(g_e)_\# \mu_e$. Using first the common map g_a , Theorem 2.2 gives

$$W_1((g_a)_\# \mu_a, (g_a)_\# \mu_b) = \frac{2 \sinh \eta}{k M_a}. \quad (9)$$

The bounds (7) follow immediately from $\|D_e\|_{\text{op}} \leq 1$.

Moment matching yields

$$\begin{aligned} |M_a - M_b| &= \left| \sum_{m=k}^{\infty} \frac{\eta^m}{m!} \int x^m d(\mu_a - \mu_b)(x) \right| \\ &\leq 2 \sum_{m=k}^{\infty} \frac{\eta^m}{m!} \leq 2e^{\eta_{\text{sep}}} \frac{\eta^k}{k!}. \end{aligned} \quad (10)$$

Coupling the same $X \sim \mu_b$ under g_a and g_b gives

$$W_1((g_a)_\# \mu_b, (g_b)_\# \mu_b) \leq \frac{|M_a - M_b|}{M_a} \leq 2e^{2\eta_{\text{sep}}} \frac{\eta^k}{k!}. \quad (11)$$

The main term in (9) is at least $2e^{-\eta_{\text{sep}}} \eta/k$, whereas the ratio of (11) to this quantity is at most

$$e^{3\eta_{\text{sep}}} \frac{\eta^{k-1}}{(k-1)!} \leq e^{3\eta_{\text{sep}}} \eta_{\text{sep}}.$$

Choose η_{sep} so that the last expression is at most $1/2$. The triangle inequality and (8) prove (6). $\square$

3 The Poissonized Schur likelihood

Let $N > 0$. At endpoint e , draw

$$Q \sim \text{Poi}(NM_e),$$

and, conditional on $Q = q$, give the observer q copies of a Haar-random conjugate of σ_e , together with the value of q . Let Γ_e denote the averaged direct-sum state of this variable-copy experiment.

For a partition $\lambda \vdash q$ with at most d rows, let f^λ be the dimension of the corresponding irreducible S_q -module, and let s_λ be the Schur polynomial.

Proposition 3.1 (Exact Poissonized Schur likelihood). *The probability of observing shape λ at endpoint e is*

$$P_e(\lambda) = e^{-NM_e} \frac{N^{|\lambda|}}{d^{|\lambda|} |\lambda|!} f^\lambda s_\lambda(A_e). \quad (12)$$

For the maximally mixed reference law P_0 ,

$$\frac{P_e(\lambda)}{P_0(\lambda)} = e^{-N(M_e-1)} \frac{s_\lambda(e^{\eta D_e})}{s_\lambda(\mathbb{K}_d)}. \quad (13)$$

Furthermore,

$$D_{\text{tr}}(\Gamma_a, \Gamma_b) = \text{TV}(P_a, P_b). \quad (14)$$

Proof. Schur–Weyl duality gives

$$(\mathbb{C}^d)^{\otimes q} \cong \bigoplus_{\substack{\lambda \vdash q \\ \ell(\lambda) \leq d}} \mathcal{Q}_\lambda^{(d)} \otimes \mathcal{P}_\lambda.$$

Since $\rho^{\otimes q}$ commutes with the S_q action, its λ block has the form $X_\lambda(\rho) \otimes I_{\mathcal{P}_\lambda}$ and has trace $f^\lambda s_\lambda(\rho)$. Haar twirling replaces $X_\lambda(\rho)$ by a scalar multiple of the identity. Hence the conditional state inside each block is maximally mixed and independent of ρ ; only the block weight $f^\lambda s_\lambda(\rho)$ depends on the spectrum.

Conditional on $Q = q$, the block weight is $f^\lambda s_\lambda(\sigma_e)$. Since s_λ is homogeneous of degree q ,

$$\begin{aligned} e^{-NM_e} \frac{(NM_e)^q}{q!} f^\lambda s_\lambda(\sigma_e) &= e^{-NM_e} \frac{(NM_e)^q}{q!} f^\lambda \frac{s_\lambda(A_e)}{(dM_e)^q} \\ &= e^{-NM_e} \frac{N^q}{d^q q!} f^\lambda s_\lambda(A_e), \end{aligned}$$

which proves (12). This exact cancellation of M_e^q is the reason for the endpoint-dependent Poisson mean. Dividing by the maximally mixed formula gives (13). Finally, the endpoint states are direct sums of identical normalized block states with different classical weights, so their trace distance is the total variation distance of those weights. $\square$

Remark 3.2 (Grand-canonical normalization). The choice $Q \sim \text{Poi}(NM_e)$ is not an approximation. It is engineered so that the factor M_e^Q from the Poisson law exactly cancels the factor M_e^{-Q} arising from the normalization of σ_e . What remains is the unnormalized Gibbs matrix $A_e = e^{\eta D_e}$ inside the Schur polynomial.

4 Characters as HCIZ integrals

For Hermitian $A, B \in \mathbb{C}^{d \times d}$ and $z \in \mathbb{C}$, define

$$I_d(A, B; z) := \int_{U(d)} \exp(zd \operatorname{Tr}(AUBU^*)) \, dU. \quad (15)$$

For a partition λ with at most d rows, define

$$\ell_i(\lambda) := \lambda_i + d - i, \quad L_\lambda := \operatorname{diag}(\ell_1(\lambda), \dots, \ell_d(\lambda)),$$

and let

$$L_\emptyset := \operatorname{diag}(d-1, d-2, \dots, 0).$$

For an arbitrary scalar centering $c \in \mathbb{R}$, put

$$B_{\lambda, c} := \frac{L_\lambda - c\mathbb{K}_d}{\sqrt{N}}, \quad u := \frac{\eta\sqrt{N}}{d}. \quad (16)$$

We use

$$\Delta(z_1, \dots, z_d) := \prod_{1 \leq i < j \leq d} (z_i - z_j).$$

Proposition 4.1 (Exact character–HCIZ ratio). *Let D be traceless Hermitian. Then, for every partition λ with at most d rows and every $c \in \mathbb{R}$,*

$$\boxed{\frac{s_\lambda(e^{\eta D})}{s_\lambda(\mathbb{K}_d)} = \frac{I_d(D, B_{\lambda, c}; u)}{I_d(D, B_{\emptyset, c}; u)}}. \quad (17)$$

Proof. Assume first that the eigenvalues $x_1, \dots, x_d$ of D are distinct, and put

$$G_d := \prod_{j=1}^{d-1} j!, \quad p := \frac{d(d-1)}{2}.$$

Weyl's character and dimension formulas give

$$\frac{s_\lambda(e^{\eta D})}{s_\lambda(\mathbb{K}_d)} = G_d \frac{\det(e^{\eta x_i \ell_j})_{i,j=1}^d}{\Delta(\ell_1, \dots, \ell_d) \det(e^{\eta x_i (d-j)})_{i,j=1}^d}. \quad (18)$$

The determinantal HCIZ formula gives

$$I_d(D, B_{\lambda, c}; u) = G_d \frac{\det(e^{udx_i(\ell_j - c)/\sqrt{N}})_{i,j=1}^d}{(ud)^p \Delta(x_1, \dots, x_d) \Delta((\ell_1 - c)/\sqrt{N}, \dots, (\ell_d - c)/\sqrt{N})}.$$

Here $ud/\sqrt{N} = \eta$, the scalar shift contributes $e^{-\eta c \operatorname{Tr} D} = 1$, and

$$(ud)^p \Delta((\ell_1 - c)/\sqrt{N}, \dots, (\ell_d - c)/\sqrt{N}) = \eta^p \Delta(\ell_1, \dots, \ell_d).$$

Thus $I_d(D, B_{\lambda, c}; u)$ is a scalar independent of λ times

$$\frac{\det(e^{\eta x_i \ell_j})_{i,j=1}^d}{\Delta(\ell_1, \dots, \ell_d)}.$$

For the empty partition $\emptyset$, one has $\ell_j = d - j$ and

$$\Delta(d-1, d-2, \dots, 0) = \prod_{j=1}^{d-1} j! = G_d.$$

Taking the ratio with the same formula for $\emptyset$ therefore gives exactly (18). Repeated eigenvalues follow by continuity. The denominator in (17) is positive for real u because the defining HCIZ integrand is positive. $\square$

Combining Theorems 3.1 and 4.1, there is a deterministic scalar

$$c_{ab} := -N(M_b - M_a) + \log I_d(D_a, B_{\emptyset, c}; u) - \log I_d(D_b, B_{\emptyset, c}; u) \quad (19)$$

such that

$$\log \frac{P_b(\lambda)}{P_a(\lambda)} = c_{ab} + \log I_d(D_b, B_{\lambda, c}; u) - \log I_d(D_a, B_{\lambda, c}; u). \quad (20)$$

The entire remaining problem is to prove that the last two terms are close whenever $B_{\lambda, c}$ lies in a fixed operator-norm ball. We will never need to estimate the scalar (19) directly.

5 The analytic core: low moments determine the HCIZ jet

This section proves the implication

$$\mathrm{tr}(D_a^j) = \mathrm{tr}(D_b^j) \ (j < k) \implies \log I_d(D_b, B; u) - \log I_d(D_a, B; u) = O(d^2 \theta^k)$$

whenever B lies in a fixed operator-norm ball and the coupling is a fixed fraction of Novak's zero-free radius.

There are only three ingredients:

1. low-order HCIZ coefficients are symmetric polynomials of the corresponding degree;
2. Novak gives a dimension-independent zero-free disc;
3. two elementary one-variable lemmas turn these facts into a remainder bound.

5.1 Low moments determine the free-energy coefficients

Since $I_d(A, B; 0) = 1$, its logarithm has a unique holomorphic branch near zero that vanishes at zero. Write

$$\log I_d(A, B; z) = \sum_{m \geq 1} F_m(A; B) z^m. \quad (21)$$

Lemma 5.1 (Invariant free-energy coefficients). *Fix B and $m \geq 1$. The coefficient $F_m(A; B)$ is a conjugation-invariant homogeneous polynomial of degree m in the entries of A . Consequently, there is a polynomial $Q_{m, B}$ such that, for every Hermitian A ,*

$$F_m(A; B) = Q_{m, B}(\mathrm{Tr} A, \mathrm{Tr}(A^2), \dots, \mathrm{Tr}(A^m)). \quad (22)$$

Proof. First expand

$$I_d(A, B; z) = 1 + \sum_{j \geq 1} H_j(A; B) z^j.$$

The coefficient $H_j(A; B)$ is a conjugation-invariant homogeneous polynomial of degree j in A : this follows directly by expanding the exponential in (15) and using Haar invariance. The formal identity

$$\log(1 + w) = \sum_{r \geq 1} \frac{(-1)^{r+1}}{r} w^r$$

shows that F_m is a universal finite linear combination of products

$$H_{j_1}(A; B) \cdots H_{j_r}(A; B), \quad j_1 + \cdots + j_r = m.$$

Every such product is conjugation-invariant and homogeneous of degree m . Hence so is F_m .

On diagonal matrices $A = \text{diag}(x_1, \dots, x_d)$, the coefficient F_m is therefore a symmetric homogeneous polynomial of degree m in $x_1, \dots, x_d$. By the fundamental theorem of symmetric polynomials, it is a polynomial in the elementary symmetric functions $e_1, \dots, e_d$. Degree considerations permit only e_j with $j \leq m$, and Newton's identities express each such e_j as a polynomial in the power sums

$$p_r = x_1^r + \cdots + x_d^r = \text{Tr}(A^r), \quad r \leq j.$$

This proves (22). □

Corollary 5.2 (Equality of the low-order free-energy jets). *Suppose*

$$\text{tr}(A^j) = \text{tr}(A'^j), \quad 1 \leq j < k.$$

Then, for every fixed B ,

$$\log I_d(A, B; z) - \log I_d(A', B; z) = O(z^k) \quad (z \rightarrow 0). \quad (23)$$

Proof. The two matrices have the same dimension, so equality of normalized traces is equality of the corresponding unnormalized traces. By Theorem 5.1, their coefficients F_m in (21) agree for every $m < k$. □

This is the complete low-order cancellation argument. No moment expansion, Weingarten formula, or separate passage from the raw HCIZ integral to its logarithm is required.

5.2 Stable nonvanishing gives the free-energy bound

We use only the nonvanishing part of Novak's theorem, not the topological expansion.

Theorem 5.3 (Novak's stable nonvanishing theorem). *There is an absolute constant $r_0 > 0$ such that, for every d and all Hermitian contractions A, B ,*

$$I_d(A, B; z) \neq 0 \quad \text{whenever } |z| \leq r_0.$$

Source and normalization. In Novak's notation,

$$I_N^{(2)}(A, B; z) = \int_{U(N)} e^{zN \text{Tr}(AUBU^{-1})} dU,$$

which is exactly (15) with $N = d$ and $U^{-1} = U^*$. His polydisc $D_N(\varepsilon)$ imposes $|z| \leq \varepsilon$ and places every eigenvalue of both external fields in the unit disc. The stable nonvanishing assertion is contained explicitly in [2, Theorem 4.1]. Restricting to Hermitian contractions gives the stated theorem. □

The following elementary half-disc lemma is a convenient form of the Borel–Carathéodory theorem.

Lemma 5.4 (Half-disc lemma). *Let f be holomorphic on a neighborhood of $|z| \leq R$, with $f(0) = 0$, and suppose that $M > 0$ and*

$$\Re f(z) \leq M \quad (|z| \leq R).$$

Then

$$|f(z)| \leq 2M \quad (|z| \leq R/2).$$

Proof. Set

$$g(z) := \frac{f(z)}{2M - f(z)}.$$

The inequality $\Re f \leq M$ implies

$$|2M - f|^2 - |f|^2 = 4M(M - \Re f) \geq 0,$$

so $|g| \leq 1$ and $g(0) = 0$. Schwarz’s lemma gives $|g(z)| \leq |z|/R$. Hence $|g| \leq 1/2$ on the half-disc, and

$$f = \frac{2Mg}{1 + g}$$

has modulus at most $2M$ there. □

Corollary 5.5 (Uniform HCIZ free-energy bound). *Put*

$$\rho := \frac{r_0}{2}.$$

For every d and all Hermitian contractions A, B , the branch

$$\Phi_d(A, B; z) := d^{-2} \log I_d(A, B; z), \quad \Phi_d(A, B; 0) = 0,$$

is holomorphic on $|z| \leq r_0$ and satisfies

$$|\Phi_d(A, B; z)| \leq 2r_0, \quad |z| \leq \rho. \tag{24}$$

Proof. Nonvanishing gives the logarithm branch. Since A, B are contractions,

$$|\mathrm{Tr}(AUBU^*)| \leq \|A\|_2 \|B\|_2 \leq d.$$

Therefore, for $|z| \leq r_0$,

$$|I_d(A, B; z)| \leq \int e^{|z|d|\mathrm{Tr}(AUBU^*)|} \mathrm{d}U \leq e^{r_0 d^2}.$$

Consequently

$$\Re \log I_d(A, B; z) = \log |I_d(A, B; z)| \leq r_0 d^2.$$

Apply Theorem 5.4 to $f(z) = \log I_d(A, B; z)$ with $M = r_0 d^2$, then divide by d^2 . □

5.3 The high-order cancellation estimate

Theorem 5.6 (High-order HCIZ cancellation). *Suppose D_a, D_b are Hermitian contractions satisfying*

$$\mathrm{tr}(D_a^j) = \mathrm{tr}(D_b^j), \quad 1 \leq j < k.$$

Let B be Hermitian with $\|B\|_{\mathrm{op}} \leq R$. If

$$uR \leq \theta\rho$$

for some $0 < \theta < 1$, then

$$|\log I_d(D_b, B; u) - \log I_d(D_a, B; u)| \leq 4r_0 d^2 \theta^k. \quad (25)$$

Proof. The case $R = 0$ is trivial. Otherwise put $\tilde{B} = B/R$ and consider

$$F(z) := d^{-2}(\log I_d(D_b, \tilde{B}; z) - \log I_d(D_a, \tilde{B}; z)).$$

By Corollary 5.5, F is holomorphic on $|z| \leq \rho$ and satisfies $|F| \leq 4r_0$ there. By Corollary 5.2, it has a zero of order at least k at zero. Hence $F(z)/z^k$ is holomorphic, and the maximum-modulus principle gives

$$|F(z)| \leq 4r_0 \left(\frac{|z|}{\rho} \right)^k, \quad |z| \leq \rho.$$

Take $z = uR$ and multiply by d^2 . □

Remark 5.7 (Where $k \asymp \log d$ enters). At a fixed coupling ratio $\theta < 1$, the nonconstant log-likelihood is bounded by $O(d^2 \theta^k)$. Making this small requires $k = \Theta(\log d)$. This d^2 is the natural free-energy scale; the proof no longer invokes any detailed expansion of that free energy.

6 A fixed-probability Schur–Weyl window

The analytic theorem applies when

$$B_{\lambda, c_N} = \frac{L_\lambda - c_N \mathbb{K}_d}{\sqrt{N}}$$

has bounded operator norm. We need only a fixed good-set probability, so second moments suffice.

For a sorted probability vector $p = (p_1, \dots, p_d)$ and $q \geq 0$, let $\mathrm{SW}^q(p)$ denote the Schur–Weyl distribution

$$\mathbb{P}_{\lambda \sim \mathrm{SW}^q(p)}[\lambda] = f^\lambda s_\lambda(p).$$

Equivalently, it is the RSK shape of a length- q i.i.d. word with letter distribution p .

Theorem 6.1 (Individual-row mean-square estimate). *There is an absolute constant C_{SW} such that, for every sorted probability vector p , every $q \geq 0$, and every $i \in [d]$,*

$$\mathbb{E}_{\lambda \sim \mathrm{SW}^q(p)} (\lambda_i - p_i q)^2 \leq C_{\mathrm{SW}} \min\{1, dp_i\} q \leq C_{\mathrm{SW}} q. \quad (26)$$

Source. This is [3, Theorem 1.5]; the case $q = 0$ is immediate. □

Fix

$$\tau := \frac{1}{100}. \quad (27)$$

Choose an absolute constant $K \geq 1$ so large that

$$\frac{2 + 4C_{\text{SW}}}{K^2} \leq \tau, \quad (28)$$

and put

$$R := 2K + 3. \quad (29)$$

Choose

$$0 < u_0 \leq \min \left\{ 1, \frac{\rho}{4R} \right\}, \quad (30)$$

and then choose a global perturbation threshold

$$0 < \eta_0 \leq \min \left\{ \eta_{\text{sep}}, \frac{1}{10}, u_0 \right\}. \quad (31)$$

For $0 < \eta \leq \eta_0$, set

$$N := \frac{u_0^2 d^2}{\eta^2}, \quad c_N := \frac{N}{d} + \frac{d-1}{2}. \quad (32)$$

We now take the centering in (16) to be $c = c_N$. Then

$$u = \frac{\eta\sqrt{N}}{d} = u_0. \quad (33)$$

Proposition 6.2 (Uniform row window). *For either endpoint $e \in \{a, b\}$ and all sufficiently large d ,*

$$P_e \left[\max_i |\ell_i - c_N| > R\sqrt{N} \right] \leq \tau. \quad (34)$$

On the complementary event,

$$\|B_{\lambda, c_N}\|_{\text{op}} \leq R, \quad uR \leq \frac{\rho}{4}.$$

Proof. Fix an endpoint. Write the ordered eigenvalues of D_e as $y_1 \geq \dots \geq y_d$, and put

$$\gamma_i := e^{\eta y_i}, \quad p_i := \frac{\gamma_i}{dM_e}, \quad \mu := NM_e.$$

Then $Q \sim \text{Poi}(\mu)$, and conditional on $Q = q$ the shape has law $\text{SW}^q(p)$.

Since $\text{Var } Q = \mu \leq e^{\eta_0} N \leq 2N$, Chebyshev's inequality gives

$$\mathbb{P}[|Q - \mu| > K\sqrt{N}] \leq \frac{2}{K^2}. \quad (35)$$

On the complementary event, $Q \leq 2N$ for all sufficiently large d . Moreover, for every i ,

$$\left| p_i Q - \frac{N}{d} \right| \leq \frac{\gamma_i}{dM_e} |Q - NM_e| + \frac{N}{d} |\gamma_i - 1|. \quad (36)$$

For $\eta \leq 1/10$, one has $\gamma_i/M_e \leq e^{2\eta_0} \leq 2$ and $|\gamma_i - 1| \leq 2\eta$. Hence, on the event in (35),

$$\left| p_i Q - \frac{N}{d} \right| \leq \left(\frac{2K}{d} + 2u_0 \right) \sqrt{N} \leq (K + 2u_0) \sqrt{N}. \quad (37)$$

Conditional on any $q \leq 2N$, Theorem 6.1 and Chebyshev's inequality give, for $i = 1, d$,

$$\mathbb{P} \left[|\lambda_i - p_i q| > K\sqrt{N} \mid Q = q \right] \leq \frac{2C_{\text{SW}}}{K^2}.$$

A union bound over the top and bottom rows, together with (35), shows that with probability at least

$$1 - \frac{2 + 4C_{\text{SW}}}{K^2} \geq 1 - \tau,$$

one has simultaneously

$$\left| \lambda_1 - \frac{N}{d} \right|, \left| \lambda_d - \frac{N}{d} \right| \leq (2K + 2u_0)\sqrt{N}.$$

Since $\lambda_1 \geq \lambda_i \geq \lambda_d$, the same bound holds for every row.

Finally,

$$\ell_i - c_N = \left(\lambda_i - \frac{N}{d} \right) + \left(\frac{d+1}{2} - i \right),$$

and

$$\frac{d-1}{2\sqrt{N}} \leq \frac{\eta}{2u_0} \leq \frac{1}{2}.$$

Because $u_0 \leq 1$, the resulting bound is at most $(2K + 3)\sqrt{N} = R\sqrt{N}$. This proves (34). The coupling bound follows from (30) and (33). $\square$

7 Likelihood normalization and the testing contradiction

We now prove Theorem 1.1 for an arbitrary ambient dimension D . Choose k to be the least even integer at least

$$A_0 \log D, \tag{38}$$

where $A_0 > 1/\log 2$ is a sufficiently large absolute constant, and set

$$d := k \left\lfloor \frac{D}{k} \right\rfloor. \tag{39}$$

For all sufficiently large D ,

$$\frac{D}{2} \leq d \leq D, \quad k \mid d, \quad k \leq \frac{d}{2}, \quad k = \Theta(\log D).$$

We run the preceding construction in dimension d . By Theorem 2.1, any lower bound obtained there also holds in dimension D .

Define

$$\mathcal{G} := \left\{ \lambda : \max_i |\ell_i(\lambda) - c_N| \leq R\sqrt{N} \right\}. \tag{40}$$

By Theorem 6.2,

$$P_a(\mathcal{G}^c), P_b(\mathcal{G}^c) \leq \tau. \tag{41}$$

On $\mathcal{G}$, Theorem 5.6 with $\theta = 1/4$ gives

$$\log \frac{P_b(\lambda)}{P_a(\lambda)} = c_{ab} + r_\lambda, \quad |r_\lambda| \leq \delta_D := 4r_0 d^2 4^{-k}. \tag{42}$$

By the choice of A_0 , one has $\delta_D = o(1)$.

The scalar c_{ab} is eliminated by normalization: if both distributions put almost all their mass on a set where their likelihood ratio is almost constant, that constant must be close to one.

Lemma 7.1 (Normalizing an almost-constant likelihood). *Let P_a, P_b be probability distributions and suppose*

$$P_a(\mathcal{G}), P_b(\mathcal{G}) \geq 1 - \tau', \quad \log \frac{P_b}{P_a} = c + r, \quad |r| \leq \delta \quad \text{on } \mathcal{G},$$

where $0 \leq \tau', \delta \leq 1/10$. Then

$$\text{TV}(P_a, P_b) \leq 2\delta + 3\tau'. \quad (43)$$

Proof. Integrating the likelihood ratio over $\mathcal{G}$ gives

$$e^{c-\delta} P_a(\mathcal{G}) \leq P_b(\mathcal{G}) \leq e^{c+\delta} P_a(\mathcal{G}).$$

Since both good-set masses lie in $[1 - \tau', 1]$ and $-\log(1 - \tau') \leq 2\tau'$, it follows that

$$|c| \leq \delta + 2\tau'.$$

Hence, on $\mathcal{G}$,

$$\left| \log \frac{P_b}{P_a} \right| \leq 2\delta + 2\tau' \leq \frac{2}{5}.$$

For $|x| \leq 2/5$, one has $|e^x - 1| \leq 2|x|$, and therefore

$$\left| \frac{P_b}{P_a} - 1 \right| \leq 4(\delta + \tau') \quad \text{on } \mathcal{G}.$$

Thus

$$\begin{aligned} \text{TV}(P_a, P_b) &\leq \frac{1}{2} \sum_{\lambda \in \mathcal{G}} P_a(\lambda) \left| \frac{P_b(\lambda)}{P_a(\lambda)} - 1 \right| + \frac{1}{2} (P_a(\mathcal{G}^c) + P_b(\mathcal{G}^c)) \\ &\leq 2(\delta + \tau') + \tau'. \end{aligned}$$

□

For all sufficiently large D , $\delta_D \leq 10^{-2} = \tau$. Therefore

$$D_{\text{tr}}(\Gamma_a, \Gamma_b) = \text{TV}(P_a, P_b) \leq 2\delta_D + 3\tau < \frac{1}{10}. \quad (44)$$

We now choose the perturbation size. Let c_0 be from Theorem 2.3, and set

$$\eta := C_0 k \varepsilon, \quad (45)$$

where $C_0 \geq 4/c_0$. Then

$$d_{\text{TV}}^{\perp}(\text{spec } \sigma_a, \text{spec } \sigma_b) \geq 4\varepsilon. \quad (46)$$

The condition $\eta \leq \eta_0$ holds whenever $\varepsilon \leq c/\log D$ for a sufficiently small absolute c .

By (32),

$$N = \frac{u_0^2 d^2}{C_0^2 k^2 \varepsilon^2} = \Theta\left(\frac{D^2}{\varepsilon^2 \log^2 D}\right). \quad (47)$$

At endpoint e , the copy count has mean $NM_e \geq e^{-\eta_0} N$. Set

$$n := \left\lfloor \frac{e^{-\eta_0}}{2} N \right\rfloor. \quad (48)$$

A Poisson lower-tail bound gives

$$\mathbb{P}[Q < n] \leq e^{-c_1 N} \quad (49)$$

uniformly at both endpoints.

Suppose there were an n -copy estimator in dimension d with sorted-TV error at most ε and worst-case success at least $2/3$. In the variable-copy experiment, whenever $Q \geq n$, discard all but n copies, run the estimator, and output the endpoint spectrum closer to its estimate. By (46), an ε -accurate estimate identifies the endpoint. The resulting test succeeds with probability at least $2/3 - o(1)$.

The Helstrom bound and (44), however, limit equal-prior discrimination success to

$$\frac{1}{2} (1 + D_{\text{tr}}(\Gamma_a, \Gamma_b)) < \frac{11}{20}.$$

This contradiction proves $N_{\text{spec}}(d, \varepsilon) > n$. Finally, Theorem 2.1 and (47) give

$$N_{\text{spec}}(D, \varepsilon) \geq N_{\text{spec}}(d, \varepsilon) = \Omega\left(\frac{D^2}{\varepsilon^2 \log^2 D}\right).$$

Renaming D as d proves Theorem 1.1.

8 Perspective and open problems

8.1 What is doing the work

The durable mechanism is

moment matching $\implies$ high-order zero of an invariant free energy $\implies$ likelihood flatness.

The Chebyshev construction supplies nearly extremal Wasserstein separation under exact bounded-support moment matching. Haar randomization forces the experiment to depend on conjugation invariants. Low-order invariant theory directly identifies the free-energy Taylor jet, while stable nonvanishing controls the entire remainder. The Young-diagram window is only a localization lemma ensuring that the random second external field remains in the analytic domain.

8.2 Which parts required the main innovations

The endpoint-dependent Poissonization is the decisive algebraic design choice: it turns the normalized Gibbs states into unnormalized exponentials inside the Schur likelihood. The HCIZ step is the deeper analytic choice: it identifies a free energy whose low-order jet is governed by the matched moments and whose full analytic size is only $O(d^2)$. The latter bound follows from stable nonvanishing and an elementary half-disc argument. The row-window argument is deliberately only a second-moment estimate: stronger concentration would add machinery without strengthening the lower bound.

8.3 Why the logarithmic loss is natural for this construction

The first logarithm is forced by the free-energy scale:

$$d^2 \theta^k \ll 1 \implies k = \Theta(\log d).$$

The second comes from approximation theory. Degree- $(k-1)$ moment matching on a bounded interval permits Wasserstein separation only $O(1/k)$, and the Chebyshev pair attains that order. Thus spectral separation ε requires $\eta \asymp k\varepsilon$, while $N \asymp d^2/\eta^2$. Removing the $\log^2 d$ denominator likely requires a different construction—for example approximate moment matching tailored to the actual HCIZ coefficients, a multiscale family, or cancellation beyond the uniform Schwarz bound.

8.4 The likely piecewise quantum problem

The classical rate (2) cautions against expecting one formula in all regimes. The present lower bound is naturally a quantum bias branch. It remains open whether there is a separate quantum variance branch, where its phase transition lies, and how the two mechanisms interact. Constant accuracy is also unresolved: the present perturbative construction requires $\eta \asymp \varepsilon \log d$ to remain small.

8.5 Further questions

1. Can one formulate a reusable HCIZ–Schur contiguity theorem: moment matching plus a bounded Schur–Weyl window implies indistinguishability whenever $d^2\theta^k$ is small?
2. Can a nonperturbative construction yield a near-quadratic lower bound at constant accuracy?
3. Is there a second hard family corresponding to the classical variance term $\sqrt{d/n} \wedge n^{-1/3}$?
4. Can local or approximate moment matching improve the logarithmic factors?
5. Which other unitarily invariant functionals admit the same free-energy mechanism?

Remark 8.1 (Why superquadratic copy counts cause no difficulty). The coupling remains fixed:

$$u = \frac{\eta\sqrt{N}}{d} = u_0.$$

As ε decreases, N grows, but the normalized shifted-row field stays in the same fixed operator-norm ball. Neither the zero-free theorem nor the Schur–Weyl window imposes $N \leq d^2$.

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